

The Subject Of Everchanging

When the world changes so much against you, will you take the time to fight back?

This module will initially consist of a stage counter, a wider component area, and a timer behind the wider component area. To disarm the module, take note of all the stages as well as active components that are shown on the module. Calculate the value needed to input from the individual sections provided. Then, perform the correct action ranging from entering digits on a keypad or inputting directions of a forgotten arrow module. The module may randomly request an input early if none of the smaller components are revealed, so bare in mind when accounting for that.

Inputting a calculated value incorrectly or letting the timer run out when the module requests an input while multiple solves are being processed will result in a strike.

The smaller components range from a digit on the display, an LED, and a cube, which may rotate in a random direction.

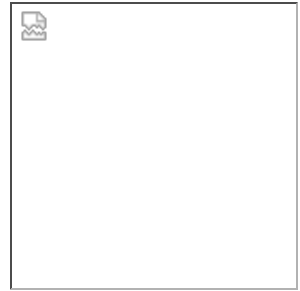


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Stage Generation Procedures

Override: Boss Mode

In this mode, each stage will be generated based on the number of non-ignored modules detected. The timer will count down at 15 seconds per stage that is currently shown and advance to the next stage when the timer expires and a non-ignored module is solved. By default, smaller components are randomly selected to be active or not active. However there are rules that can be applied to alter how the stages are generated, which may be overridden by missions or settings. By default, this mode will be active if the module is able to obtain an ignore list.

These rules apply in boss mode with the 1st rule having highest priority:

- If there are 6 or fewer non-ignored unsolved modules left, the given stages from the non-ignored modules must reveal at least 1 of the smaller components. Likewise, the next stage immediately after those will ALWAYS be the input procedure.
- The first 3 stages at the start will reveal exactly 1 of the smaller components.
- If none of the smaller components are revealed for that given stage, the module will then request an early input. The next 6 stages generated after the current stage must reveal at least 1 of the smaller components. In the case of multiple solves being processed, a stage requiring early inputs will have a timer set for 2 minutes instead of 15 seconds. The timer may be extended by 15 seconds for every stage after the 8th requested stage.

Override: Exhibition Mode

To distinguish between exhibition mode and boss mode, the timer will be fully disabled in exhibition mode as well as the display on the timer saying "EXHIBITION." In this mode, the defuser is able to view each stage at their own pace and can access the next/previous stage by pressing the right/left half of the timer. As a consequence, there are no stages that require early inputs. This mode may be enforced in the settings if necessary.

The only other rule this mode enforces is if the current stage no. is equal to the number of solvable modules present, that given stage must reveal at least 1 of the smaller components. Likewise, the next stage immediately after that will ALWAYS be the input procedure.

Individual Stage Calculations: General Overview

There's so much to prep for.

In order to obtain the calculated value, the expert will need to find the page which only contains the specified components shown on that stage and refer to page on how to retrieve the calculated value. If colorblind mode is enabled for Everchanging, a letter will also be shown to denote the color of the LED if it's on, or no letters at all if it's off. This letter is the first letter of the given color, I.E. R for Red, O for Orange, etc. If the digit is present on any of the stages, the name of the color will be shown to the right of the digit, if the stage requires the color of the digit or requires disambiguation.

The components are only active if they are visible during that stage.

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Individual Stage Calculations: Singular LED

If the smaller active components are not [LED], you are looking at the wrong page! [Click here to go back to General Overview for Individual Stage Calculations.](#)

The LED will show exactly 1 color which can be one of the following colors in the table provided. If colorblind mode is enabled for Everchanging, a letter will also be shown to denote the color of the LED if it's on, or no letters at all if it's off. This letter is the first letter of the given color, I.E. R for Red, O for Orange, etc.

The calculated value is determined on the current stage number shown on the module and the color of the LED, from the cell in the given table provided.

Stage no. % 2	LED Color									
	White	Red	Orange	Yellow	Green	Lime	Cyan	Blue	Purple	Magenta
1	0	1	2	3	4	5	6	7	8	9
0	5	6	7	8	9	0	1	2	3	4

Do the dice look similar to the ones from that one switch party game?

The cube will show a random digit from 0 to 9 on each face. Each cube shown on this manual corresponds to a calculated value, however the digit shown on the module is represented by dots on a given domino side. 0 will be shown as literally nothing on the given face. The cube will also rotate randomly to show each face if possible.

Possible Nets

1	Net 0			1	Net 1			1	Net 2			0	Net 3		
2	3	5	4	3	3	3	5	1	1	4	4	2	4	4	4
6				6				9				6			
0	Net 4			2	Net 5			0	Net 6			0	Net 7		
1	3	3	5	2	2	2	6	4	4	4	4	3	3	3	3
7				6				4				8			
3	Net 8			1	Net 9										
3	3	3	4	1	1	6	6								
4				6											

Individual Stage Calculations: Lonely Digit

If the smaller active components are not [Digit], you are looking at the wrong page! [Click here to go back to General Overview for Individual Stage Calculations.](#)

The digit shown will show a random number from 0 - 9 on the module when it is revealed. The digit may also show a random color which can be safely ignored as the color has no relevance for this section.

Start with the digit currently on the display. Modify it by this table underneath in respect to the conditions:

Condition	Modifier
Serial number contains a vowel	-1
Batteries	+1 for each
The last digit of the serial number is even	+1
Parallel port & Lit NSA present	Undo those modifications

Then keep adding or subtracting 10 after the using the table provided, until you have a number with 0 - 9 inclusive. This your calculated value for this stage.

Individual Stage Calculations: Simple LED Math

If the smaller active components are not [LED, Digit], you are looking at the wrong page! [Click here to go back to General Overview for Individual Stage Calculations.](#)

A digit and LED will show denoting a possible expression made by the color of the digit, the LED, and the number shown on the digit. The digit will be colored in white or green while the LED will be in one of 10 different colors, including off. If colorblind mode is enabled for Everchanging, a letter will also be shown to denote the color of the LED if it's on, or no letters at all if it's off; likewise, the color of the digit will also be shown to the right of the digit that is currently shown.

The calculated value is determined by evaluating this expression: (digit shown) (digit color)(LED color) and then adding or subtracting 10 continuously until the result is within 0 - 9 inclusive. Do bare in mind (digit color) and (LED color) need to be replaced with specific operators and values respectively!

LED color		Value
White		4
Red		9
Orange		0
Yellow		8
Green		1
Lime		7
Cyan		2
Blue		6
Purple		5
Magenta		3

Color of Digit	Operator
White	+
Green	-

Individual Stage Calculations: AlphaForget

To be fair, he saw a harder variant of this and didn't approve that difficulty. At all.

If the smaller active components are not [LED, Cube], you are looking at the wrong page! [Click here to go back to General Overview for Individual Stage Calculations.](#)

An LED will light up alongside a cube transforming into 8 separate spheres, and performing a specific rotation. Determine the calculated value by plugging in the function for that given rotation. The diagram shown here will show the axis for each rotation specified. Note that +Y is shown on the top as if the defuser is facing at the module but the diagram shows all 3 possible axes.

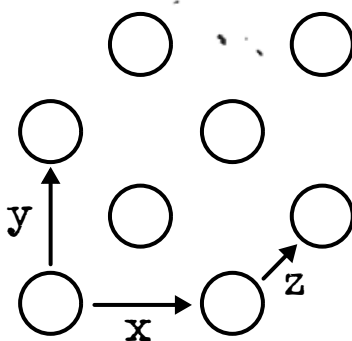


Figure Not To Scale

LED Color -> LED Value									
White	Red	Orange	Yellow	Green	Lime	Cyan	Blue	Purple	Magenta
7	4	1	8	5	2	9	6	3	0

For the rotation table provided, D_n refers to the $(n+1)$ th digit in the serial number; S refers to the stage number the module is currently on when this combination of smaller components were revealed; and L is the LED value of the wider table provided above for that stage. If CV is not within the range of 0-9, repeatedly add or subtract 10 until it is.

Rotation Shown	Function To Apply
YX	$CV = L + S$
XY	$CV = S - L$
ZX	$CV = D_0 + S$
XZ	$CV = D_0 - S$
YZ	$CV = L + D_1$
ZY	$CV = D_1 - L$

Individual Stage Calculations: Cube Desynchronization

Are you forgetting what perspective you should follow instead?

If the smaller active components are not [Cube, Digit], you are looking at the wrong page! [Click here to go back to General Overview for Individual Stage Calculations.](#)

A random digit will be shown and a random rotation will be applied to the cube. The current displayed digit is then modified by the rotation shown, and then a hidden value is added to the result of this rotation. This new value, modulo 10, is then displayed and used as the next digit for the next random rotation. The hidden digit is the calculated value for this stage. This entire procedure repeats until another stage is requested. The diagram show the axis of the rotations shown where the Y+ axis is where the defuser is looking directly on top of the module.

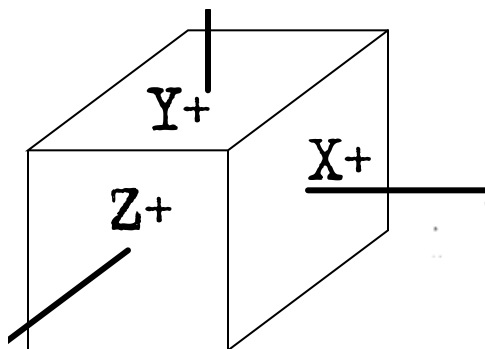


Figure Not To Scale

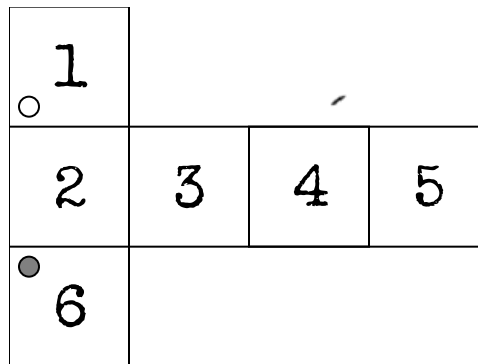
To determine the hidden value, apply the last displayed digit to the last rotation shown based on the table provided to the right. Now subtract the digit obtained from this table from the current digit displayed, adding 10 if the result is negative. This results in your calculated value for this stage.

Given Rotation	Last Digit									
	0	1	2	3	4	5	6	7	8	9
X (X+ - axis CW)	5	8	4	1	0	3	9	2	6	7
X' (X+ - axis CCW)	1	4	7	0	8	9	6	3	5	2
Y (Y+ - axis CW)	4	2	8	9	6	5	3	0	7	1
Y' (Y+ - axis CCW)	3	9	0	8	7	1	4	5	2	6
Z (Z+ - axis CW)	7	3	2	6	5	0	1	8	9	4

Individual Stage Calculations: The Cube Lite

If the smaller active components are not [LED, Cube, Digit], you are looking at the wrong page! [Click here to go back to General Overview for Individual Stage Calculations.](#)

The LED will show a random color out of red, orange, white, green, blue, and purple; the digit will cycle between 2 different digits; and the cube will show 6 digits on the module from 0 - 9 inclusive, with the first and last faces being marked with an LED. To distinguish between the first and last faces with the LED, the first face will have the LED on the bottom left in white and the last face will have the LED on the top left in dark gray. The faces are labeled in the net shown on this page here, as if the faces are right-side up.



The first table in this page provide the LED value for this stage.

Red	No. of Modules (including needles) + 5	Orange	Stage No.	White	2
Green	6 + Sum of Serial No. Digits	Blue	1 + (number of letters in Serial No.)	Purple	Sum of digits on the cube

The cube will also perform 5 rotations. These rotations are determined from a top-down perspective. One of these rotations is distinct from the 5 rotations and is converted into a given value from the table provided.

Rotate CCW	2nd face digit	Rotate CW	5th face digit	Tilt Up	1st face digit
Tilt Down	6th face digit	Tip Right	3rd face digit	Tip Left	4th face digit

Add the LED value with the value of the provided rotation and then modulo 8 if the current stage number is $10n + 5$, 9 if the current stage number is $10n + 6$, or 10 otherwise. Add that result with the sum of the digits that are shown, and then modulo 10. This results in your calculated value for that stage.

Input Procedures: General Overview

When the module successfully reveals no components or is ready to solve, the module will reveal one of the two input methods that the defuser can interact with. Each of the provided pages underneath will provide different procedures that are required to disarm the module. If any inputs were incorrect for that stage, the module may reshew that given stage that was inputted incorrectly.

The order of the calculated values is when each stage was shown and excluding previous correct sets if there are any.

If the defuser manages to input all of the requested stages correctly when the module successfully reveals none of the smaller components, the module will clear all of the requested stages up to this input procedure and speed up the timer if the timer has not expired for this stage. Those stages that were inputted correctly will NOT be included in the next input procedure. If this is the final input procedure, the module will instead disarm itself instead.

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2. [Monochrome Arrows](#)

Input Procedures: 10 Digit Keypad

It's going to Forget Me... Not. Now? Maybe... Later? No.

If the revealed component does not contain [Keypad], you are looking at the wrong page! [Click here to go back to General Overview for Input Procedures.](#)

The keypad will show 10 digits ranging from 0-9 respectively, with one of the keys missing a label, alongside the number of digits to input to complete this input procedure. The key with the missing label will always input the 10th digit out of the rest of the displayed labels.

Start with the least significant digit of the calculated value from each stage that requests it. Where the left-most column is the amount of digits inputted so far, the new digit for each stage will be calculated value modified by the first condition from the given set of conditions, modulo 10 (kept within 0-9 by adding/subtracting 10).

0	FRK indicator present	+2 if lit FRK, -2 otherwise
	No lit indicators	-(unlit indicators)
	# lit > # unlit indicators	-7
	Otherwise	+(S.N. 2nd digit)
1	Empty port plate present	0
	Prev. input % 2 != 0	+(# of stages to input % 10)
	Otherwise	-(1st digit inputted - 1)
2+	Last 2 prev. inputs contain a 0	+(S.N. 1st digit)
	Last 2 prev. inputs both odd	+(sum of all odd digits in S.N.)
	Otherwise	+(left-most digit sum of 2 prev. inputs)

Subtract this digit from (10 + the digit with the missing label). If this result is over 9, take the last digit. This results in the digit you must input for that stage.

Input Procedures: Monochrome Arrows

Looks like someone made the sequel of Black Arrows again. This time, it wants more than just grayness on a hexagonal board.

If the revealed component does not contain [Arrows], you are looking at the wrong page! [Click here to go back to General Overview for Input Procedures.](#)

This module will reveal 4 arrows, with 4 different shades of gray on the module. To complete this input procedure, input the arrows based on the characteristics shown. Inputting the incorrect arrow will reshuffle each stage used for this input procedure.

To obtain the characteristics, you will need to use the calculated values obtained earlier. Add all of the calculated values by how many times the input procedure has activated, including this input procedure, and modulo them by 10. The 3×3 table will show the directions to move relative to the starting position.

Starting on the top-left of the 8×8 grid, move 1 space to the right for every "9", and 1 space down for every "0". This is now your starting location. Then, for each non 0/9 number in order, move relative to the 3×3 table on the right. This 8×8 grid wraps around horizontally and vertically.

1	2	3
4		5
6	7	8

E	F	C	B	A	H	D	G
A	D	H	F	B	G	E	C
F	G	E	D	H	A	C	B
G	H	B	E	F	C	A	D
H	B	G	C	E	D	F	A
B	A	D	H	C	F	G	E
C	E	A	G	D	B	H	F
D	C	F	A	G	E	B	H

Each letter obtained from the starting position, every time the direction changes from the current position (I.E moving TL, then BL), and from the end of the sequence, will correspond to characteristic from an arrow that should be pressed. In a case that no movement is performed, only the letter at the starting position is noted. Press them in the order obtained. Up, Down, Left, Right are all abbreviated as U, D, L, R, respectively. Shades of gray refer to the color of that arrow.

Letter	A	B	C	D	E	F	G	H
Characteristic					U	R	D	L